

University of Sindh – IMCS

HCI and Computer Graphics

Assignment #1

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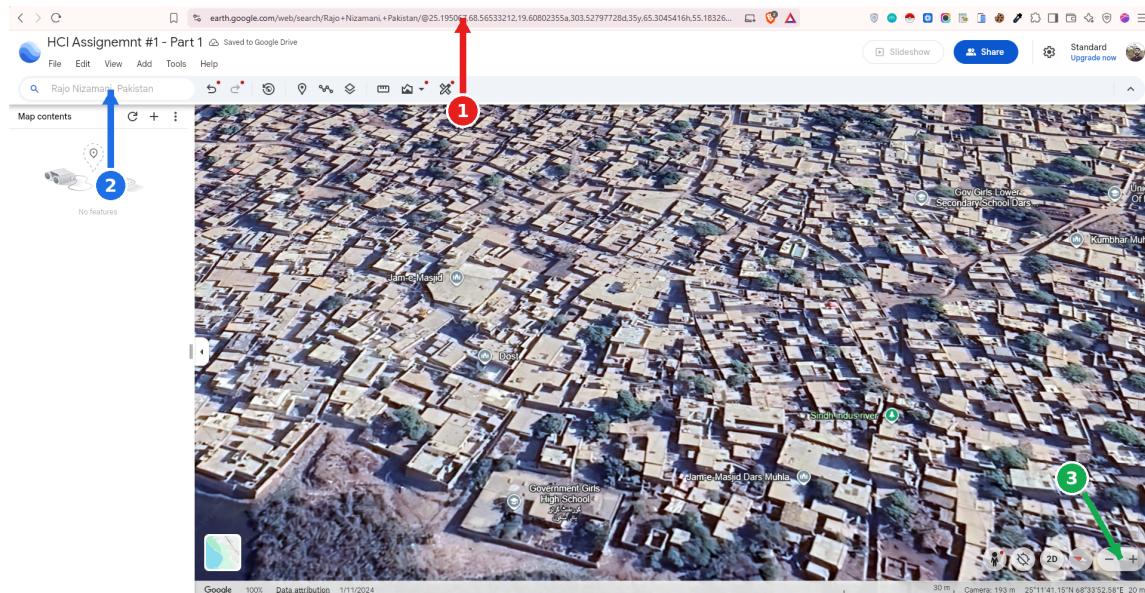
Semester: 6th

Subject: HCI and Computer Graphics

Submitted To: Sir Rajesh

HCI - Assignment #1: 3D Map App Overview — Part #1

This assignment part focuses on an interface analysis of Google Earth, as 3D Map Overview.

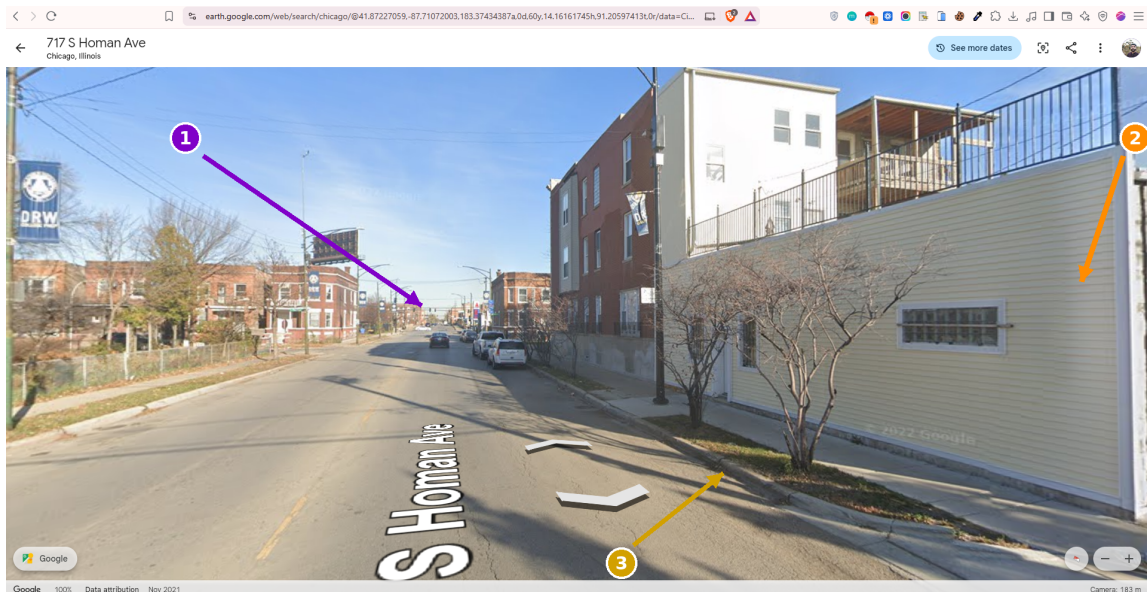


- **Blue Arrow (search bar):** The blue arrow points to Google Earth's in-app search field — the actual input point where a user types a location to search.
- **Red Arrow (URL bar):** The red arrow points from the map view up to the browser's address bar, showing how the URL updates live to reflect the current search ("Rajo Nizamani") — a form of system feedback.
- **Zoom Controls (Green Arrow):** The green arrow points to the bottom-right corner of the interface, highlighting the "+" (zoom in) and "-" (zoom out) controls. These allow users to adjust the map scale for detailed inspection or broader geographical context.

This is the first part of the assignment.

HCI - Assignment #1: 3D Map App — Part 2 (CG Focus)

This part examines the computer graphics techniques of Google Earth's Street View.



- **Perspective Camera View (Purple Arrow):** The purple arrow points down S Homan Ave to where the road and buildings converge, showing perspective projection — distant objects shrink, creating the illusion of depth.
- **Texture Mapping (Orange Arrow):** The orange arrow points to the plain siding on the beige building's exterior, where photographic material imagery is mapped onto its 3D surface. This gives the flat building geometry a realistic, detailed appearance.
- **Light Source & Shadows (Gold Arrow):** The gold arrow points to the shadow cast on the sidewalk by nearby trees/structures, implying a single light source (the sun) and helping ground objects in 3D space.

This is the second part of the assignment.

HCI - Assignment #1: Reflection Questions — Part #3

Question 1: How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

Answer: The realistic satellite imagery and 3D view in Google Earth actually make it a lot easier to recognize real-world places and get a feel for how everything is laid out, so navigating feels pretty natural. That said, the tilted 3D angle isn't always helpful — it can make text labels harder to read and distances tricky to judge, since the perspective kind of warps the scale of things.

Question 2: What is one simple HCI change you would make to improve the user interface?

Answer: I'd make the zoom controls more obvious — right now they're just small icon buttons that are easy to overlook. Adding a visible "+/-" label next to them would make it clearer at a glance what they do, especially for someone using the app for the first time.

This is the third part of the assignment.